Moderator & TA Troubleshooting Guide

Tiny Tapeout Workshop Series

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For TAs and helpers circulating the room.

Quick lookup of every place participants commonly get stuck, organized by activity. Companion to the Speaker Script.

Speaker Script = what the moderator says & does.
This document = what the helpers do when something breaks.

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Part I

Beginner Session

Friday April 17, 2026 Tech L440 5:00–9:00 PM

1 Level 1: SiliWiz

If stuck

- If someone can't find the layer selector: it's the colored squares on the left side of SiliWiz.
- If the simulation view is blank: make sure all three contacts are placed and touching the right regions.
- If someone is done early: ask them to try building a CMOS inverter (NMOS + PMOS with drains connected).

2 Level 2: Wokwi (Build Your Own Gate)

If stuck

- "Input pin not driven" warning = a chip input is floating. Connect unused inputs to VCC or GND.
- If someone wired it wrong: most common mistake is wrong pin numbers.
- Fast finishers: try building a half adder (XOR + AND) or chain two full adders.

3 Level 3: GitHub Actions

3.1 Linking the Wokwi Project

If stuck

The single most common failure: **wrong `wokwi_id`**. It must be the number from the URL, not the project name. If the GDS action fails, check this first.

3.2 Other GHA failures

If stuck

- Red `gds` action — check the Actions log. Most common: wrong `wokwi_id`, design too large, floating port in the Wokwi schematic.
- Red `docs` action — usually missing fields in `info.yaml` or malformed `docs/info.md`.
- GitHub Pages 404 — Settings → Pages was not switched to "GitHub Actions" as the source.

4 Level 4: VGA Playground / Verilog

If stuck

"What Is This Code?" is the densest slide in the workshop. Go slow. Point at each line on screen as the moderator explains it. If people look lost, reassure them: *"You do not need to memorize this. The goal is just to know enough to change one line."*

If stuck

- Coloured underlines in VGA Playground = syntax error. The screen freezes on the last valid state until it's fixed.
- GHA red after editing `project.v` = usually the module name in the file does not match `top_module` in `info.yaml`.
- Forgot to add `hvsync_generator.v` to `PROJECT_SOURCES` in `test/Makefile` = synthesis fails on undefined module.

5 Level 5: OSS CAD Suite (Local Install)

If stuck

- **Mac (Apple Silicon):** after `tar -xf`, the binary may be quarantined — System Settings → Privacy & Security, click "Allow Anyway".
- **Windows:** use the `.exe` installer, then run `oss-cad-suite\environment.bat` in every new `cmd`.
- `yosys: command not found` = they forgot to `source oss-cad-suite/environment` in the current shell.
- `gtkwave` opens blank = they did not expand `tb` → `dut` and "Append" the signals. Then Shift+Alt+F to zoom to fit.

Part II

Advanced Session

Friday April 10, 2026 Tech L440 5:00–9:00 PM

6 Docker / Container Setup

Docker troubleshooting

- **Mac (Sonoma+):** `brew install -cask docker && open -a Docker`. Wait for the whale icon.
- **Mac (older):** Docker Desktop won't install. Use colima: `brew install colima docker && colima start`, then `export DOCKER_HOST=unix://$HOME/.colima/default/docker.sock`
- **Windows (WSL2):** Docker Desktop must be installed *and running* (whale icon in taskbar). In Docker Desktop Settings → Resources → WSL Integration, enable your distro. If `docker` is not found inside WSL, restart the WSL terminal. If Docker Desktop won't start: check that virtualization is enabled in BIOS and run `wsl -update` in PowerShell (admin).
- **Linux:** `sudo systemctl start docker`. Permission denied? `sudo usermod -aG docker $USER` and re-login.
- **XQuartz:** must be installed, logged out/in, Settings → Security → allow network clients, quit and reopen.
- `xhost` fails: try `xhost +si:localuser:$USER`
- **Nuclear option:** skip X11 entirely. Use `./start_vnc.sh` and open `http://localhost:80` in a browser.

7 Part 4: XSCHEM & NGSPICE

7.1 Simulation Failures

If stuck

- **"Model not found"** = missing `corner.sym` block in the testbench. Go back and add it with `corner=tt`.
- **Flat line at zero** = the Vdd source isn't connected, or its `value` isn't set.
- **No waveform file** = Simulation → "Set netlist dir" was not pointed at the project folder.
- **NGSPICE hangs** = pulse parameters too tight. Loosen the rise/fall edges.

8 Part 5: Magic Layout

If stuck

Layout is the hardest part of the night. TAs should be actively walking the room. The most common mistakes:

- **Poly not crossing diff** = no transistor formed.
- **Contacts not inside diff** = open circuit.
- **Wires crossing without intent** = short circuit.
- **DRC: poly too narrow** = widen to 0.15 μm minimum.
- **DRC: missing well taps** = add a well tap inside the nwell connected to vdd.
- **LVS: "Netlists do not match"** = check pin labels, source/drain orientation, and that all

contacts are placed.